

Aniruddha Chowdhury, Ph.D., EIT (P.E. passed)

Mechanical Engineer | HVAC & Refrigeration | Building Systems Design
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Summary

Mechanical Engineer with experience in mechanical design, HVAC system analysis, engineering documentation, and construction-support activities. Skilled in developing HVAC design documents using AutoCAD and Revit MEP, performing energy analysis, system auditing, interpreting ASHRAE standards, and supporting projects from concept development through implementation.

Core Competencies

HVAC Engineering: HVAC System Design, Heating and Cooling load Analysis, HVAC equipment selection, Energy analysis, Building performance evaluation, Sustainable HVAC design strategies.

Design Documentation: HVAC Construction Drawings, Mechanical Plans and Sections, Construction Documents, Equipment Schedules, CAD Drafting Standards, Shop Drawing Review, Technical Submittals.

Software: AutoCAD, Revit MEP, SolidWorks, Microsoft Office Suite, Excel, MATLAB, SQL, Power BI.

Code and Standards: ASHRAE 90.1, ASHRAE 62.1, ASHRAE 55, NFPA 820, NFPA 90A, ISO 9001, ISO 14001.

Project Management: Construction Administration, Contractor Coordination, RFI Response, Client Communication, Technical Documentation, Multidisciplinary coordination.

Professional Experience

Researcher, Louisiana Tech University, Ruston, LA | 03/2021 – 05/2026

- Directed collaborative research activities across cross-functional teams for a multi-year federally funded project.
- Developed and validated multiscale thermal models predicting temperature-dependent material behavior across a wide operating range, applying heat-transfer and thermodynamic principles central to HVAC load analysis.
- Built simulation workflows (LAMMPS, Python) to model complex physical systems and extract engineering-relevant performance metrics from large datasets.
- Translated modeling outputs into a screening tool adopted by industry partners, turning technical simulation into practical engineering recommendations.

Process Engineer, Al Razi Chemical Complex Limited, Bangladesh | 01/2018 – 12/2020

- Modeled a hydrogen peroxide production process in Aspen Plus to perform material and energy balance analysis.
- Received a Kaizen Award for improving process capability of a distillation unit (Cpk from ~1.14 to ~1.4).
- Supervised 25+ operators and oversaw standard operating procedures, and safe chemical handling practices.
- Conducted process and equipment audits, tracked KPIs, and reported findings to senior engineers.

Assistant Mechanical Engineer, AFTAB Automobiles Limited, Bangladesh | 08/2016 – 12/2017

- Developed HVAC design concepts for an automotive manufacturing facility requiring temperature control, ventilation, and process exhaust management.
- Designed HVAC zoning strategies for production areas, offices, and specialized manufacturing spaces.
- Utilized AutoCAD and SolidWorks for the design and modeling of vehicle chassis and custom components.
- Translated client specifications into technical Bill of Materials (BOM) for raw materials, authored tender bidding documents, and prepared detailed financial expense reports for clients.

HVAC Design Project for Residential Apartment Building (ASHRAE Standard 169, Zone 1A)

- Modeled a three-story, 45,000 ft² residential building using Carrier HAP v6 for heating and cooling load calculations.
- Evaluated building envelope characteristics, occupancy schedules, internal heat gains, and ventilation requirements based on ASHRAE standards.
- Prepared quarterly reports for funding agency and effectively communicated project reports to clients.
- Selected residential HVAC equipment including heat pumps, air handlers, and duct systems based on calculated loads and efficiency requirements.
- Performed duct sizing, airflow calculations, and mechanical coordination for residential spaces.

Education

Ph.D. in Engineering | Louisiana Tech University | CGPA: 3.85/4.00

Master's in Mechanical Engineering | Louisiana Tech University | CGPA: 3.92/4.00

Bachelor's in Mechanical Engineering | National Institute of Technology, Jamshedpur, India | CGPA: 3.49/4.00

Professional Certificates & License

- PE passed, (Mechanical- HVAC & Refrigeration)
- Engineer in Training (Louisiana) EI.0036277
- Six Sigma Green Belt
- Lean Manufacturing Certification
- Project Management Professional (PMP) Training